Akshu Singhal

Dehradun, India | +91-6397286592 | akshusinghal5@gmail.com | Leet-code | Github | Linkedin

Profile Summary

Final-year B.Tech (CSE) student with strong foundation in Data Structures, Algorithms, DBMS, Operating Systems, and OOP. Hands-on experience in developing Al-based Face Recognition System and Deadlock Detection System using Python and MySQL. Solved 200+ DSA problems on Leet-Code. Seeking entry-level Software Developer role.

Skills

C, C++, JAVA, Python, HTML, CSS, JavaScript, MySQL, Data Structures, Algorithms, OOP, DBMS, OS, CN Git, GitHub, VS Code, Leadership, Event Management, Public Speaking, Time Management, Communication Skills, Problem Solving

Education

Bachelor of Technology in Computer Science and Engineering (CSE) — **CGPA - 7.5**

Graphic Era Deemed University , Dehradun — **01-2022 - 01-2026**

- Courses: Object-Oriented Programming (OOP), Computer Network and Database

12th (Senior Secondary Examination, CBSE) — **Percentage - 87**

Pragyan Sthali School , Saharanpur — **01-2021 - 01-2022**

10th (Secondary Examination, CBSE) — **Percentage - 74**

Pragyan Sthali School , Saharanpur — **01-2019 - 01-2020**

Projects

Smart Attendance System using Face Recognition. — **03-2024 - 04-2024**

- Developed an Al-based Face Recognition Attendance System using Python and OpenCV, achieving 90%+ facial recognition accuracy.
- Automated attendance recording process, reducing manual errors by approximately 80%.
- Implemented real-time facial detection and recognition using NumPy and computer vision techniques.
- Integrated MySQL database for secure storage and retrieval of attendance

Notes Management System. — **08-2024 - 11-2024**

- Developed a secure web-based Notes Management System to create, edit, and manage user notes efficiently.
- Implemented User Authentication and Authorization to ensure secure login and data privacy.
- Designed and implemented full CRUD operations (Create, Read, Update, Delete) for seamless data management.
- Integrated MySQL database for structured data storage, ensuring consistency and reliability

Deadlock Dynamics (Detecting and Recovering System in Operating System). — **02-2025 - 04-2025**

- Developed a Deadlock Detection and Recovery System in Python to identify and resolve deadlock conditions in multi-process operating system environments.
- Implemented the Wait-For Graph (WFG) algorithm to detect circular wait conditions and analyse process-resource dependencies.
- Designed an automated recovery mechanism to terminate or restart affected processes, ensuring optimized system performance.
- Simulated real-time process allocation scenarios to validate deadlock detection accuracy.
- Strengthened understanding of Operating Systems concepts, including resource allocation, process synchronization, and concurrency control.

Achievements

Object-Oriented Programming in Java Certificate of Completion

- Demonstrated understanding of core OOP concepts including classes, inheritance, polymorphism, and encapsulation through practical coding exercises and assessments.

Data Structures and Algorithms using C++ Certificate of Completion

- Gained hands-on experience in implementing algorithms, optimizing time complexity, and solving real-world problems using data structures like arrays, stacks, queues, and trees.

SQL (Basic) - Hacker Rank

- Validated foundational database querying and data manipulation skills.

AWS Solutions Architecture Certificate

https://www.theforage.com/completion-certificates/pmnMSL4QiQ9JCgE3W/kkE9HyeNcw6rwCRGw_pmnMSL4QiQ9JCgE3W_6948347ffd8c77499a45ac40_1766388876399_completion_certificate.pdf